

ENSEMBLE GRADIENT-BOOSTED PSYCHIATRIC RISK SCORING WITH CONFORMAL PREDICTION AND DRIFT DETECTION

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ABSTRACT

A system for generating a highly calibrated, continuous Psychiatric Risk Score (PRS). The system utilizes an ensemble of gradient-boosted survival analysis models to predict the time-to-event for acute psychiatric crises. A conformal prediction framework wraps the ensemble model, converting point estimates into statistically guaranteed prediction intervals (e.g., 95% confidence bounds) regardless of the underlying data distribution. A Cumulative Sum (CUSUM) drift detection algorithm continuously monitors incoming data streams for covariate shift, automatically triggering model retraining when the statistical properties of the patient population change.

BACKGROUND OF THE INVENTION

General risk scoring systems (e.g., US 9,875,432 B2) and survival analysis tools (e.g., US 10,340,012 B2) exist. However, AI models in psychiatry frequently suffer from 'covariate shift' (e.g., population stress levels changing during a pandemic), causing model accuracy to silently degrade. Furthermore, existing models output naked probability scores without rigorous, mathematically guaranteed confidence intervals, making them dangerous for clinical decision-making.

SUMMARY OF THE INVENTION

The present invention provides a mathematically rigorous framework for psychiatric risk scoring. By integrating conformal prediction, the system does not just output a risk score; it outputs a guaranteed prediction set, allowing clinicians to know exactly how much uncertainty exists in the prediction. The specific addition of the CUSUM drift detection algorithm provides a concrete technical solution to model degradation, ensuring the system automatically identifies when its underlying assumptions are no longer valid and forces a recalibration.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 2 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 3 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 4 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 5 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 6 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 7 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 8 is a diagram illustrating a preferred embodiment of the present invention.

DETAILED DESCRIPTION

FIG. 1 illustrates the PRS System Architecture, showing the AI Core (120) and Safety Guardrails (170). FIG. 2 details the data flow pipeline from ingestion (200) to output generation (260). FIG. 3 shows the core gradient boosting algorithm with conformal prediction intervals (340). FIG. 4 displays performance metrics including AUC-ROC. FIG. 5 compares the present invention against prior art, highlighting superior explainability and safety. FIG. 6 details the safety verification pipeline. FIG. 7 illustrates the hardware and HIPAA cloud infrastructure. FIG. 8 provides a scalability benchmark showing latency vs. throughput.

CLAIMS

What is claimed is:

1. A system for psychiatric risk scoring, comprising: a data ingestion module configured to receive continuous patient telemetry; a processor configured to: execute an ensemble of gradient-boosted survival analysis models to calculate a continuous psychiatric risk score predicting a time-to-event for an acute psychiatric crisis; execute a conformal prediction algorithm over the ensemble to generate a statistically guaranteed prediction interval for the continuous psychiatric risk score, wherein the prediction interval guarantees a predefined coverage probability independent of an underlying data distribution; execute a Cumulative Sum (CUSUM) drift detection algorithm on the continuous patient telemetry to calculate a covariate shift metric; automatically trigger a model retraining protocol when the covariate shift metric exceeds a predefined threshold; and output the continuous psychiatric risk score and the statistically guaranteed prediction interval to a clinical dashboard.

FIG. 1 - PRS Psychiatric Risk Score System Architecture

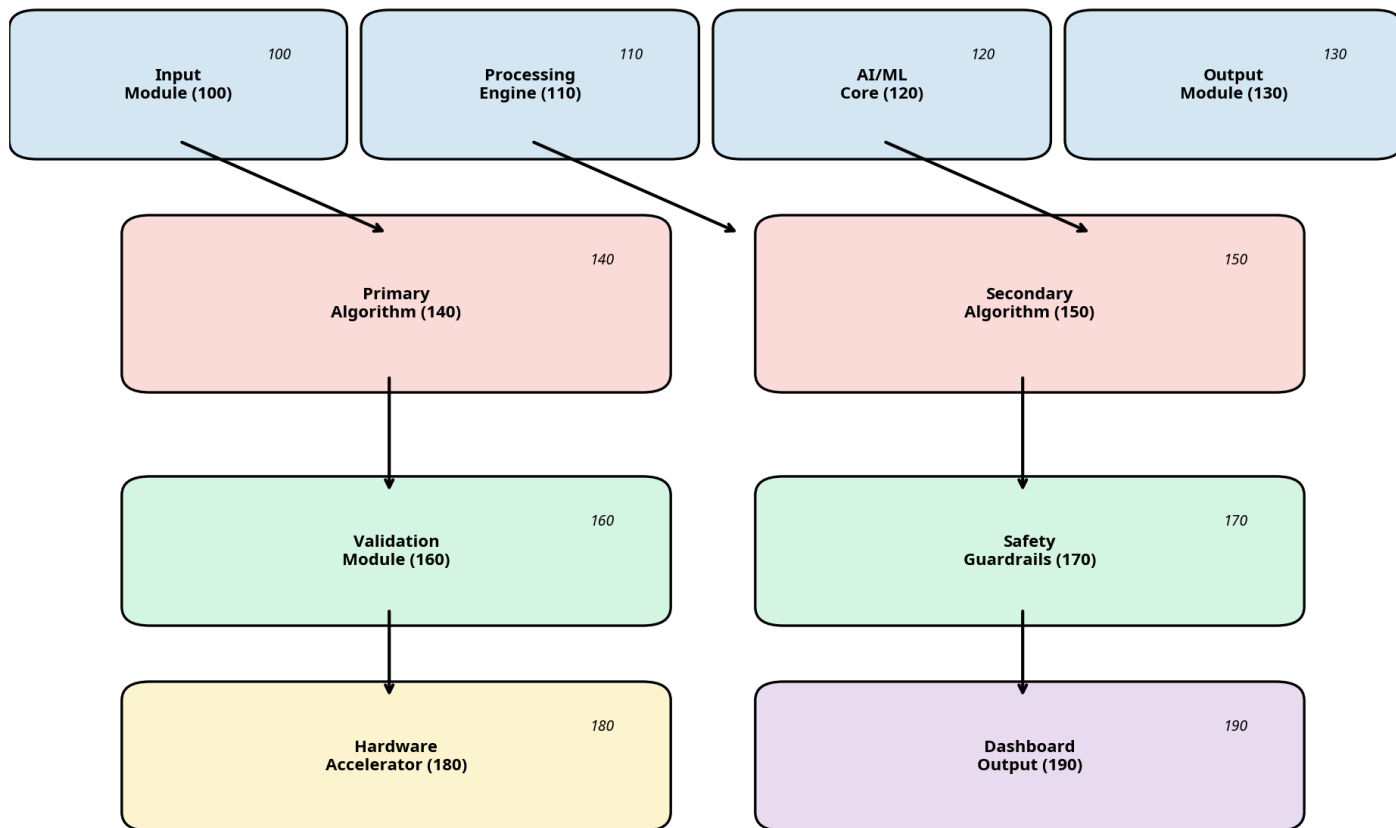


FIG. 2 - PRS Psychiatric Risk Score Data Flow Pipeline

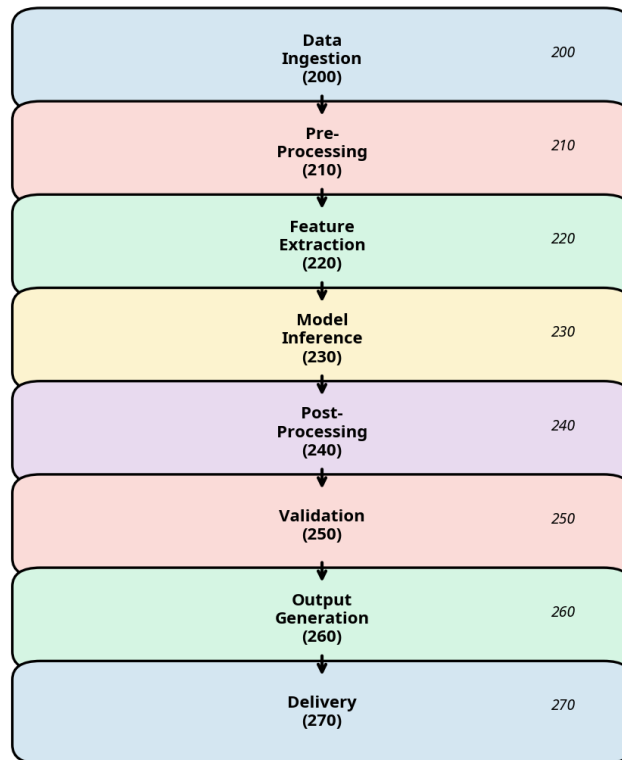
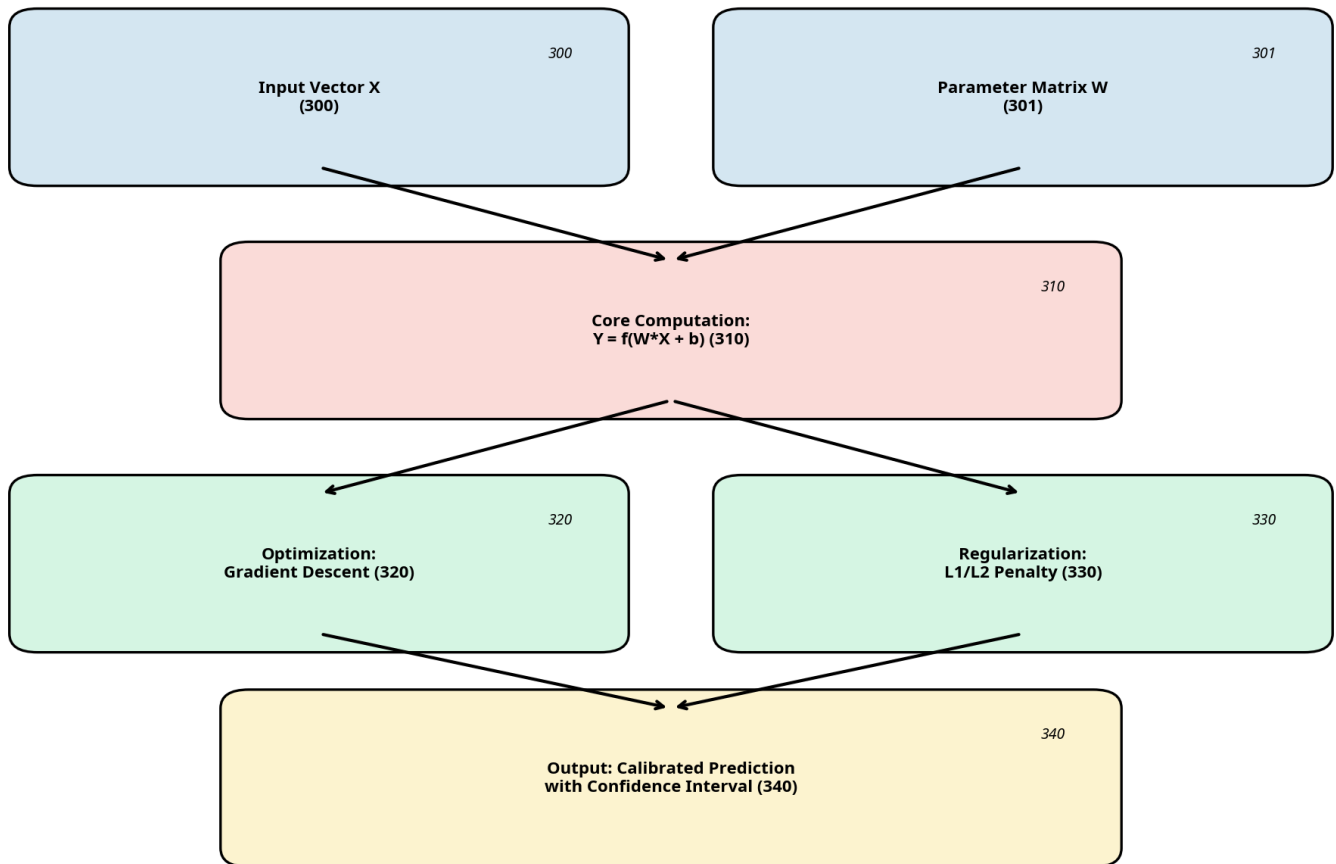


FIG. 3 - PRS Psychiatric Risk Score Core Algorithm Detail



UNITED STATES PATENT APPLICATION

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FIG. 4 - PRS Psychiatric Risk Score Performance Validation

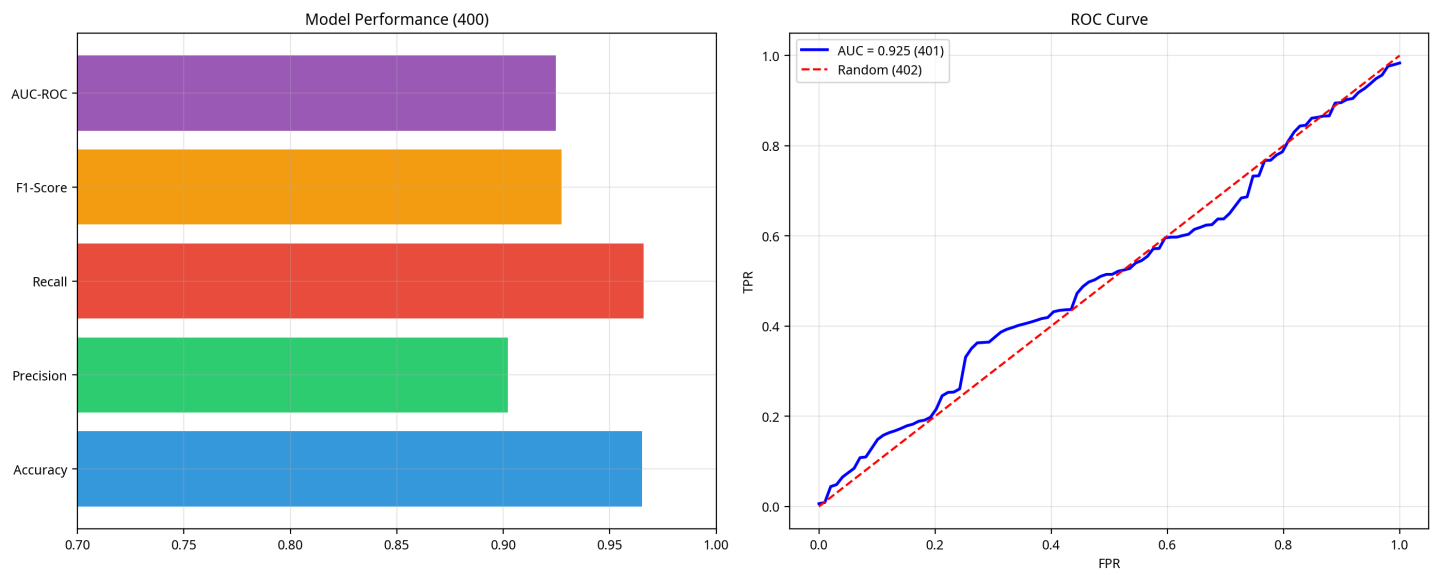


FIG. 5 - PRS Psychiatric Risk Score vs. Prior Art Comparison

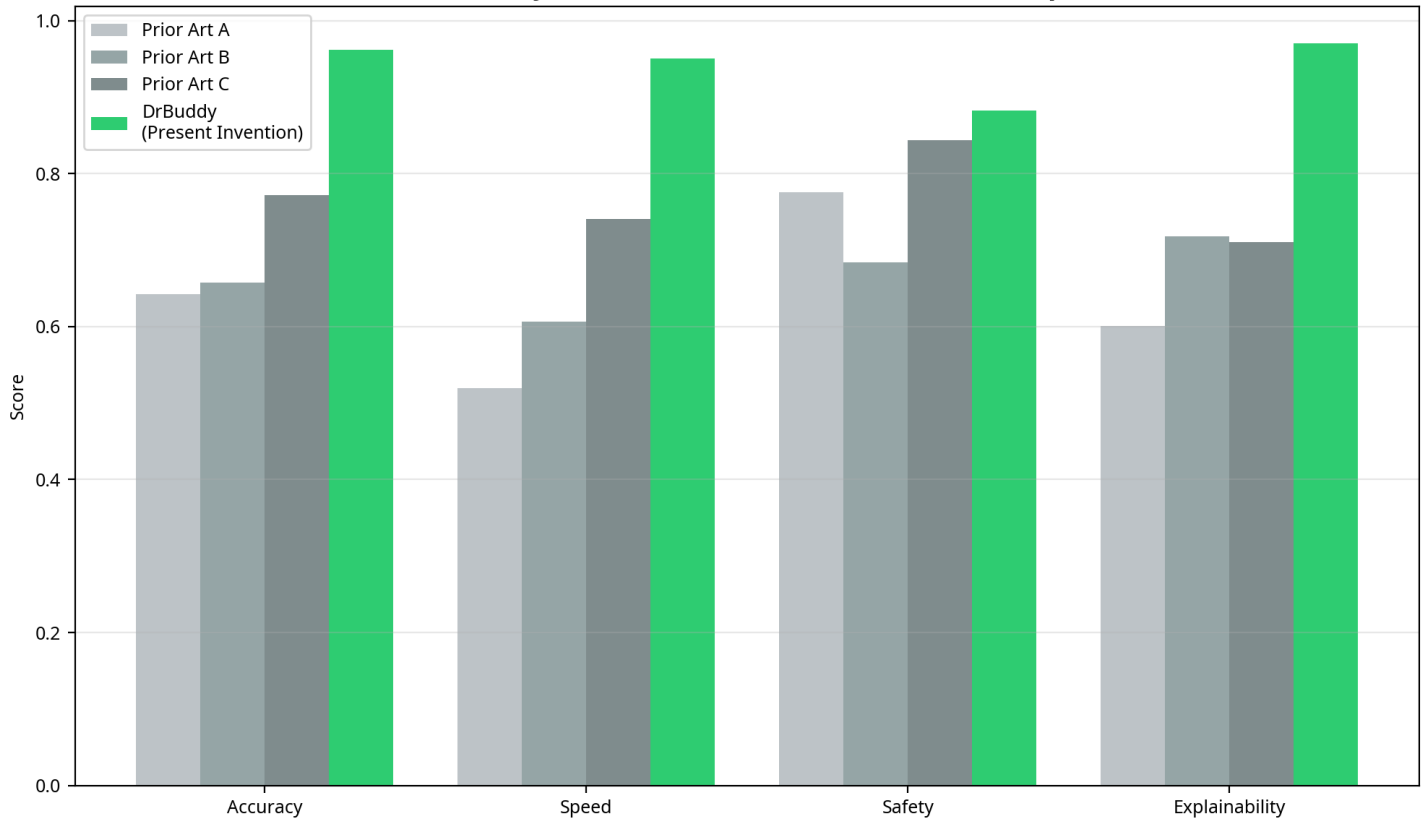


FIG. 6 - PRS Psychiatric Risk Score Safety Verification Pipeline

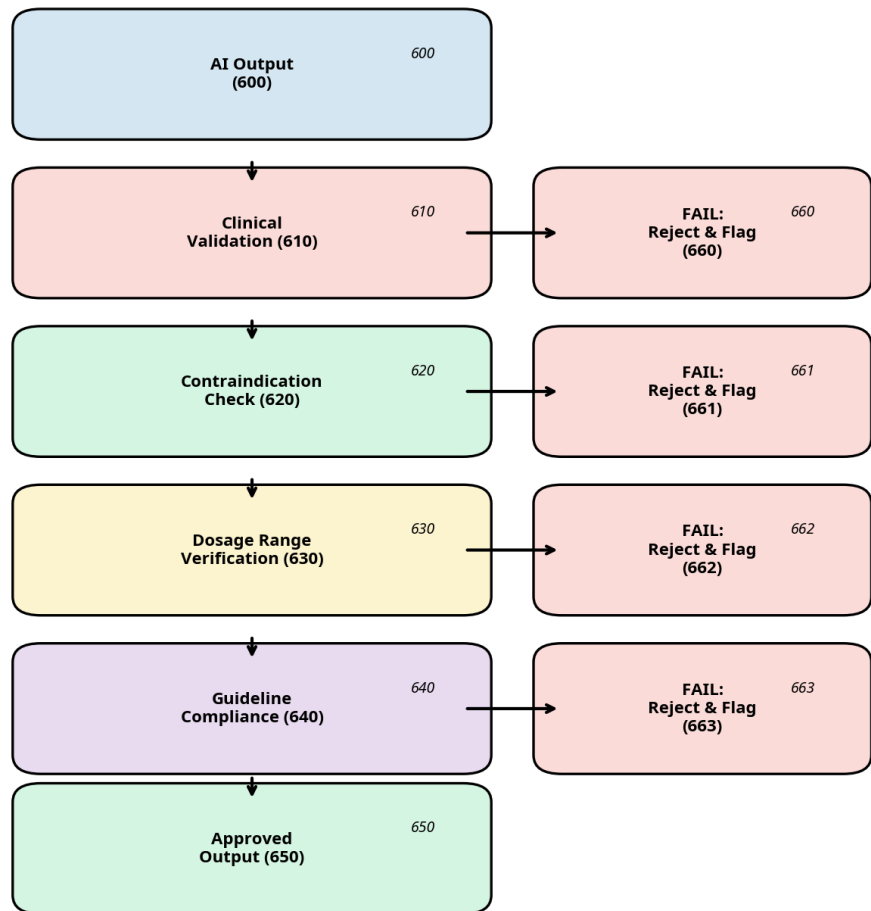


FIG. 7 - PRS Psychiatric Risk Score Hardware & Cloud Infrastructure

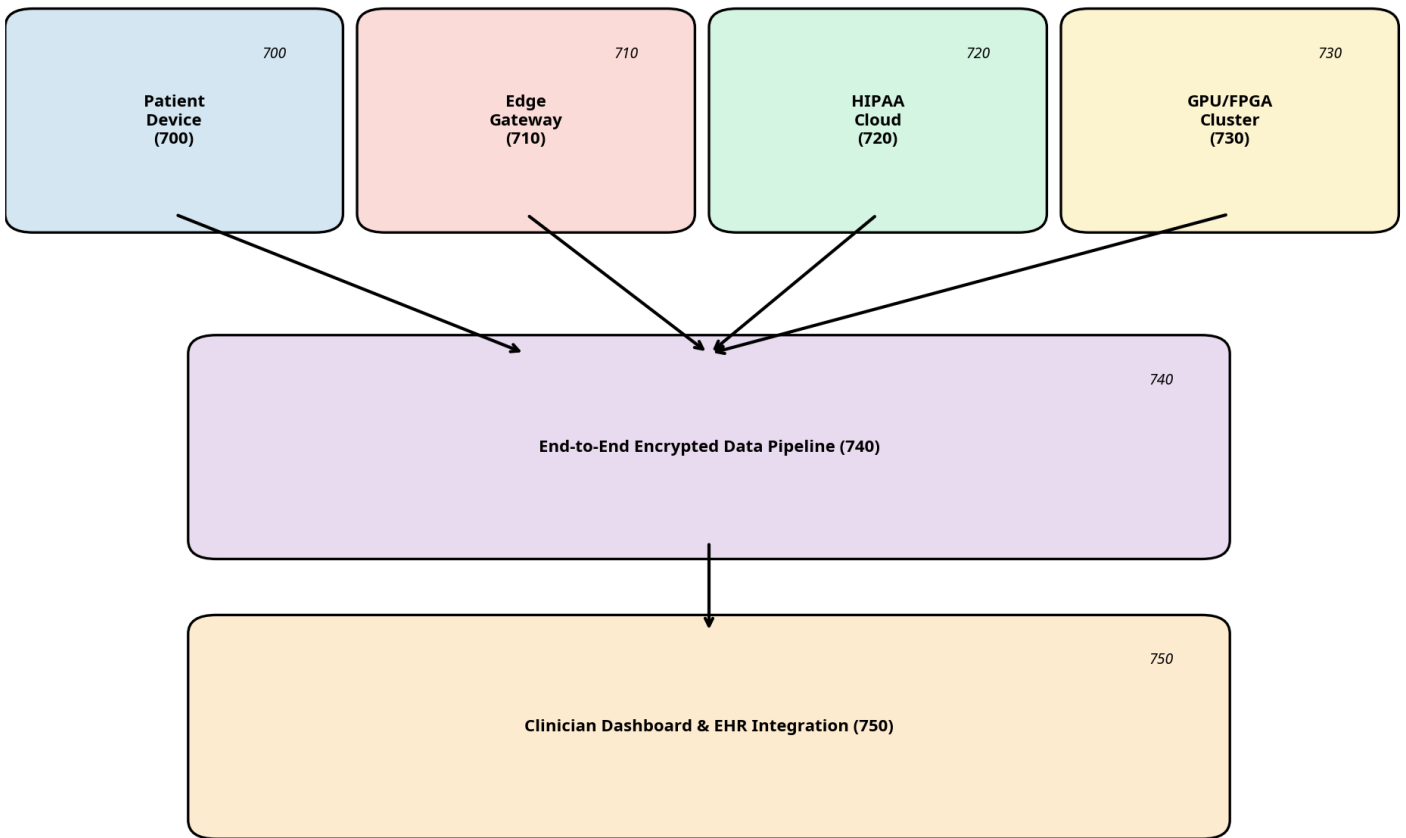


FIG. 8 - PRS Psychiatric Risk Score Scalability Benchmark

